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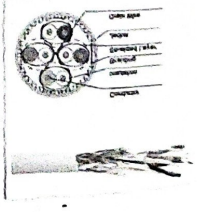
Study of Different Types of Network Cables

Aim:

To study the different types of Network Cables.

i) Understand different types of network cable.

* Unshielded Twisted Pair (UTP) cable
* Shielded Twisted Pair (STP) cable
* Coaxial Cable
* Fibre Optics cable

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use	Image
UTP	Category 3	10 bps	Advantages * Cheaper in cost. * Easy to install as they have a smaller overall diameter Disadvantage * More prone to EMI & noise.	10Base-T Ethernet Fast Ethernet, Gigabit Ethernet.	
	Category 5	Up to 100mbps			
	Category 5e	1 Gbps			
STP	Category 6, 6a	10 Gbps	Advantages * Shielded * Faster than UTP Disadvantage * Expensive * Greater installation effort.	* Gigabit Ethernet, 10G Ethernet (55mm) widely used in data centers.	 
	Category 7	10Gbps			

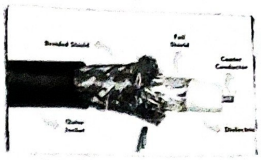
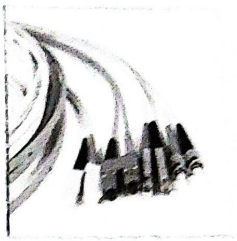
Coaxial cable — RG-6, RG-5, RG-11

Fibre optics cable — Single mode, Multi mode

b) Ma

Tools

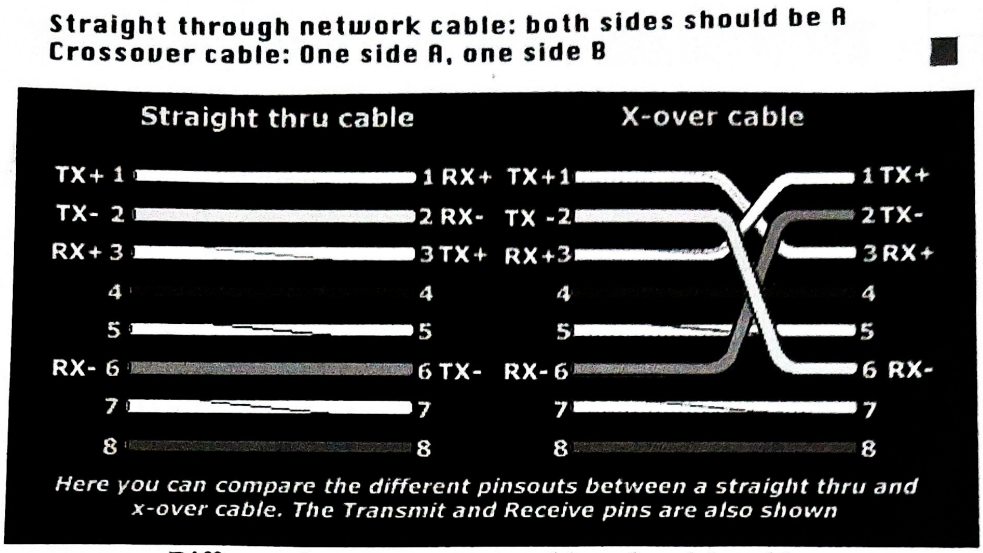
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Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	* High bandwith * Versatile Disadvantage * Cost * Size is bulky	Speed of signal is 500m Television network High speed internet connection	
fibre optics cable	Single mode Multi mode	100Gbps	Advantages *High speed * High security Disadvantages * Expensive	Maximum distance of fibre optics cable is around 100 meters.	

b) Make your own Ethernet Cross-Over Cable/Straight cable

Tools & parts needed:

* Ethernet cabling: CAT5e certified for gigabit support.
* A crimping tool: This is an all-in-one networking tool.
* Two RJ45 plugs.



Difference between crossover cable and straight cable

brown solid

white/blue stripe

Steps to construct own Ethernet :

Step 1: Start construction of the device by threading shields into the table.

Step 2: Strip ~~approx~~ approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round area.

Step 3: After, you will need to untangle the wires, there should be four ~~---~~ twisted pairs.

Steps 4: Once the order is correct, bunch them together in a line, and if there is a line.

Step 5: Push the cable right in. The notch at the end of the plug needs to be just over the cable shielding.

Step 6: After the wires are securely sitting inside the plug, insert it into the crimping tool.

Step 7: Repeat the other end using diagram.

Result:

The study of different types of Network Cables has been completed.